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PHY 103 — Past Questions & Materials

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Course: PHY 103

Institution: Osun State University, Osogbo

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Paper 1 of 1 — PHY 103 PROF. ALABI

Level 100 • First Semester • Handout

OSUN STATE UNIVERSITY, OSOGBO

FACULTY OF BASIC AND APPLIED SCIENCES.

DEPARTMENT OF PHYSICS.

PHY 103

GENERAL PHYSICS 1

COURSE LECTURER: PROF. O.O ALABI

• FLUID MECHANICS

(SOME PROPERTIES OF LIQUID: SURFACE TENSION AND CAPILLARITY)

COHESION AND ADHESION:

- Cohesion :means intermolecular force of attraction between molecules of the same liquid. Cohesion is a tendency of the liquid to remain as one assemblage of particles. Surface tension is due to cohesion between particles at the free surface.
- Adhesion :Means force of attraction between the molecules of a liquid and the molecules of a solid boundary surface in contact with the liquid. This property enables a liquid to stick to another body. Capillary action is due to both cohesion and adhesion.
- Surface Tension: It is defined as the tensile force acting in the surface of a liquid in contact with a gas or on the surface between two immiscible liquid such that the contact surface between is like a membrane and tension. Surface tension is caused by the force of cohesion as at the free surface. Some important examples of phenomenon of surface tension are rain drops, rise of the sap in the tree, collection of dust particles in water surface, break up of liquid jets. It is expressed in N/m.

• PRESSURE INSIDE A WATER DROPLET, SOAP BUBBLES AND A LIQUID JET:

4□

• Water droplet: $P = \frac{4\gamma}{r}$ -----1

□

8□

• Soap Bubble: $P = \frac{8\gamma}{r}$ -----2

□

2□

• Liquid Jet: $P = \frac{2\gamma}{r}$ -----3

□

- Where σ = surface tension of the liquid
- d = diameter
- P = pressure inside the droplet
- Example: 1: If the surface tension at air-water interface is 0.069 N/m , what is the pressure difference between inside and outside of an air bubble of diameter 0.0009 mm ?
- Solution: Lower: $\sigma = 0.069 \text{ N/m}$, $d = 0.0009 \text{ mm}$? An air bubbles has only one surface. Hence,

4

- $P = 4 \times 0.069 / 9 \times 10^{-7} = 306667 \text{ N/m}^2$

- Example 2: A soap bubble 62.5 mm diameter has an internal pressure in excess of the outside pressure of 20 N/m^2 . What is the tension in the soap film?

- Solution

- Given: Diameter of the bubble, $d = 62.5 \text{ mm} = 62.5 \times 10^{-3} \text{ m}$;

External pressure in excess of the outside pressure is, using the relation

8

- $\sigma =$

8

- $20 =$

62.5×10^{-3}

$20 \times 65.5 \times 10^{-3}$

- $\sigma = 0.156 \text{ N/m}$

8

CAPILLARITY

Capillary is a phenomenon by which liquid (depending upon its specific gravity) rises into a thin glass tube above or below its general level. This phenomenon is due to the combined effect of cohesion and adhesion of liquid particles.

i.e Rising and falling of liquid above its general level in thin glass tube. It is as a result of cohesion and adhesion.

$h = 4 \sigma \cos \theta / \rho g r$ -----4

Where d = diameter of the capillary tube

θ = angle of contact of the water surface

σ = surface tension force for unit length and

w = weight density = ρg (w needs to be properly defined for the unit to be consistent)

h = level of the liquid in the tube

For water and glass: $\theta = 0$

Hence the capillary rise of water in the glass tube,

4

$h = \text{-----} -5$

It must be noted then the smaller the diameter the greater the capillary depression.

Example 1 : A clean tube of diameter 2.5 mm is immersed in a liquid with coefficient of surface tension = 0.4 N/m. The angle of contact the liquid had with the glass can be assumed to be 135° . The density of the liquid = 13600 kg/m^3 . What would be the level of the liquid in the tube relative to the free surface of the liquid inside the tube?

Solution

Given: Diameter of the tube, $d = 2.5 \text{ mm}$, $\theta = 135^\circ$, $\rho = 13600 \text{ kg/m}^3$

$4 \cos \theta$

$h =$

$4 \times 0.4 \times \cos 135^\circ$

$h =$

$(9.81 \times 13600) \times 2.5 \times 10^{-3}$

$= -3.39 \times 10^{-3} \text{ m}$ or -3.39 mm

Negative sign indicates that there is a capillary depression (fall) of 3.39mm.

Examples 2: Determine the minimum size of glass tubing that can be used to measure water level, if the capillary rise in the tube does not exceed 0.3mm. Take surface tension of water in contact with air as 0.0735N/m.

Solution

Given: Capillary rise, $h = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$, surface tension = 0.0735N/m, specific weight of water, $w = 9810 \text{ N/m}^3$, size of glass tubing, d :

4

$h =$ (Assuming $\theta = 0$ for water)

4×0.0735

$d =$

$$0.3 \times 10^{-3} \times 9810$$

$$= 10 \text{ mm.}$$

Exercise 1: A soap bubbles 50 mm diameter has internal pressure in excess of the outside pressure of 25 N/m^2 . Calculate tension in the soap film. (Ans. 0.156 N/m)

Exercise 2: Determine the minimum size of glass tubing that can be used to measure water level, if the capillary rise in the tube does not exceed 0.25 mm . Take surface tension of water in contact with air as 0.0735 N/m . (Ans. 120 mm)

VISCOSITY

Viscosity may be defined as the property of a fluid which determines its resistance to shearing stresses. Viscosity of fluids is due to cohesion and interaction between particles.

Newton's law of Viscosity

This law states that the shear stress (τ) on a fluid element layer is directly proportional to the rate of shear strain. The constant of proportionality is called the co-efficient of viscosity.

Mathematically,

$$\tau = \eta \frac{du}{dy}$$

$$\eta = \frac{\tau}{\frac{du}{dy}} \text{-----6}$$

$$\eta = \frac{\tau}{\frac{du}{dy}}$$

Water, kerosene, and air are some of the examples of the Newtonian fluid. (Lubricant oil and water are then water)

Example 1: A plate 0.05 mm distant from a fixed plate moves at 1.2 m/s and requires a force of 2.2 N/m^2 to maintain this speed. Find the viscosity of the fluid between the plates.

Solution

Viscosity of the moving plate, $V = 1.2 \text{ m/s}$, distance between the plate, $dy = 0.05 \text{ mm} = 0.05 \times 10^{-3}$

Force on the moving plate, $F = 2.2 \text{ N/m}^2 = \tau$

$$\tau = \eta \frac{du}{dy}$$

$$\eta = \frac{\tau}{\frac{du}{dy}}$$

$$\eta = \frac{\tau}{\frac{du}{dy}}$$

$$du = \text{Change of velocity} = u - 0 = 1.2 \text{ m/s}$$

$$1.2$$

$$2.2 = \eta \times$$

$$0.05 \times 10^{-3}$$

$$2.2 \times 0.05 \times 10^{-3}$$

$$\eta =$$

$$1.2$$

$$\eta = 9.16 \times 10^{-5} \text{ Pa}\cdot\text{s}$$

Poiseuille's equation

In a steady (ordered) flow, Poiseuille states that the volume flowing per second depends on the viscosity η ,

η

the radius r of the pipe and the pressure gradient ,

η

$$\eta \propto \frac{1}{r^4}$$

$$\eta = \frac{8 \eta L \Delta p}{\pi r^4 Q}$$

$$\eta = \frac{8 \eta L \Delta p}{\pi r^4 Q}$$

Exercise: In a giving a patient a blood transfusion, the bottle is set up so that the level of blood in it is 1.3m above the needle, which has an internal diameter of 0.36mm and its 3cm in length. In one minute, 4.5cm³ of blood passes through the needle. Calculate the viscosity of blood, assuming its density is 1020kg/m³.
(Ans. $2.43 \times 10^{-3} \text{Ns/m}^2$).

INTRODUCTION TO THERMODYNAMIC PROPERTIES

The thermodynamic properties need to be considered when a fluid is influenced by change of temperature. The following equation, known as the characteristic equation of a state of a perfect gas is used for this purpose.

$$PV = MRT$$

Where, P = Absolute pressure

V = Volume of M kg of gas

M = Mass of gas

T = Absolute temperature, and

R = Characteristics gas constant

1. (a) What will be the gauge pressure and (b) the absolute pressure of water at depth 12m below the surface? $\rho_w = 1000 \text{kg/m}^3$ and $P_{\text{atmosphere}} = 101 \text{kN/m}^2$

a. $P_{\text{gauge}} = \text{density} \times \text{acceleration due to gravity} \times \text{height}$
 $= \rho gh$

$$= 1000 \times 9.8 \times 12$$

$$= 117720 \text{N/m}^2$$

$$\text{or } = 117.7 \text{kN/m}^2$$

b. Absolute pressure

$$P_{\text{absolute}} = P_{\text{gauge}} + P_{\text{atmosphere}}$$

$$= 117720 + 101000$$

$$= 218720 \text{N/m}^2 \text{ or } 218.720 \text{kN/m}^2.$$

2 (a) At what depth below the surface of oil, relative density 0.8, will produce a pressure of 120kN/m²? (b) What depth of water is the equivalent to this?

Solution

a.

Relative density of oil = 0.8

Relative density of oil = 0.8

Relative density of oil = 0.8

$\rho_o = 0.8$

$\rho_o = 0.8$

$\rho_o = 0.8$

$\rho_o = 1000 \times 0.8 = 800$

$P = \rho_o h$

120×10^3

$h = \frac{120 \times 10^3}{800} = 15.29 \text{ m}$

$1000 \times 0.8 \times 9.8$

120×10^3

b. If $\rho_w = 1000 \text{ kg/m}^3$ $h = \frac{120 \times 10^3}{1000 \times 9.8} = 12.23 \text{ m}$

$h = 12.23 \text{ m}$

3. What would pressure in KN/m² be if the equivalent head is measured as 400mm of

(a) Mercury = 13.6, (b) water (c) oil specific weight 7.9KN/m³ (d) a liquid of density 520kg/m³ ? ($g = 9.81 \text{ m/s}^2$) $h = 400 \text{ mm} = 0.4 \text{ m}$

Solution

Relative density of mercury = 13.6

a. $\rho_m = 13.6$, $\rho_w = 1000$ = relative density of water

$\rho_m = 13.6$

b. $\rho_m = 13.6 \times 1000 = 13600 \text{ kg/m}^3$

$P = \rho_m h$

$13.6 \times 10^3 \times 9.81 \times 0.4$

$= 3924 \text{ N/m}^2$

c. $\rho_o = 7.9 \text{ kN/m}^3$ $h = 0.4 \text{ m}$

$P = \rho_o h = 7.9 \times 10^3 \times 0.4$

$P = \rho_o h$

$P = 7.9 \times 10^3 \times 0.4$

3160 N/m^2

d. $\rho_l = 520 \text{ kg/m}^3$

$520 \times 9.81 \times 0.4$

$= 2040 \text{ N/m}^2$

4. A manometer connected to a pipe indicates a negative gauge pressure of 50mm of mercury. What is the absolute pressure in the pipe in newton per square meters, if the atmosphere pressure in 1 bar

(R.D. (ρ) = 13.6 for mercury).

Solution

$$h = -500 \text{ mm} = -0.05 \text{ m}$$

$$\rho_{\text{mercury}} h_{\text{mercury}} = 101325 \text{ Pa} = 1 \times 10^5 \text{ Pa} - 2$$

$$\rho_{\text{mercury}} h_{\text{mercury}} = \rho_{\text{water}} h_{\text{water}} + \rho_{\text{mercury}} h_{\text{mercury}}$$

$$= \rho_{\text{water}} h + \rho_{\text{mercury}} h$$

$$\rho_{\text{water}} h$$

$$\rho = 13.6 \text{ g/cm}^3 = 13.6 \times 1000 \text{ kg/m}^3$$

$$\rho_{\text{water}} = 1000 \text{ kg/m}^3$$

$$\rho_{\text{mercury}} h_{\text{mercury}} = ((13.6 \times 1000) \times 9.81 \times (-0.05) + 101325 \text{ Pa})$$

$$= -6670 + 101325$$

$$= 94655 \text{ Pa}$$

$$= 94.655 \text{ kPa}$$

5. What height would a water barometer need to be to measure atmosphere pressure?

Solution

$$\rho_{\text{mercury}} h_{\text{mercury}} = 101325 \text{ Pa} = 1 \times 10^5 \text{ Pa} - 2$$

$$\rho = \rho_{\text{water}} h$$

$$101325 \text{ Pa} = \rho_{\text{water}} h$$

$$101325$$

$$h = \frac{101325 \text{ Pa}}{1000 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2}$$

$$1000 \times 9.81$$

$$101325$$

$$h = 10.33 \text{ m}$$

$$= 0.75 \text{ m}$$

$$(13.6 \times 10^3 \text{ kg/m}^3) \times 9.81 \text{ m/s}^2$$